

Universal Spin-Orbit Interactions in Mie Scattering

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Abstract:

We show a widespread spin-orbit interaction (SOI) in Mie scattering that manifests itself as a four-piece scattering pattern in S3 component upon the injection of linear polarized wave. This effect is significant and even visible to naked eyes, and can bring enormous polarization effect to the scattering or diffraction from structures in Mie region. Through constructing the microscopic electrodynamic model, we strictly explain how the excited photons transfer their angular momentums to objects, leading to an elliptically rotated polarizability and differentiated torque in objects; and how the objects eliminate the polarizability, and emit photons of opposite spins to different directions. We also give a brief explanation with the multipole decomposition theory that the polarity retardation effect in object makes the scattering as the superposition of different dipoles, such as the electric and magnetic dipoles, and quadrupoles. Engineering the morphology or size of objects will optimize the combination of dipoles, and eventually provide a way to control the scattering pattern in S3 component and the underlying SOI effect.

Keywords: Photonic Spin-Hall Effect; Spin Orbital Interaction; Mie Scattering; Nano Particles; Stokes parameters

The photon spin-orbit interactions (SOI) can lead spin-splitting perpendicular to the incident plane when reflecting from an interface[1,2] or passing through an inhomogeneous medium[3,4]. This supports creative future applications using polarization degrees of freedom, such as for precision metrology[5,6], or spin-optics[7-11] akin to semiconductor spintronics. However, the experimental observation of such SOI effects usually requires ingenious but non-intuitive methods, such as ultrasensitive quantum ‘weak measurements’[12,13], due to the extremely-small Imbert-Fedorov shift [14-16] and Magnus effects[17]. This situation can be completely changed when employing spatially inhomogeneous medium with the feature size of subwavelength, where the polarization and position of lights are intrinsically coupled. The enhanced SOI together with spin-dependent phenomena is widely reported in nano-photonics[18-23], but still requires specific considerations, such as non-paraxial conditions[6,24], specifically-designed interfaces[10,25-27], and wavevector matching in spin-locked coupling via evanescent near-fields[7,20].

Mie scattering serves as a fundamental framework for interpreting optical measurements

and understanding light-matter interactions, ranging from nanoscale particles to planetary-scale atmospheric systems[28,29]. But the SOI is not yet involved in Mie scattering to date. In this work, we report a ubiquitous and significant spin-orbit interaction (SOI) in Mie scattering that doesn't require any specific consideration on observation. As shown in Figure 1(a), when a linearly-polarized plane-wave incidents on the gold NS of 80 nm along the Z- direction, the strong SOI near nano-sphere (NS) leads to a far-field scattering pattern that highly depends on both spin and orbit angular momentums, where the left- and right-hand circularly polarized light (LCP and RCP) prefer to scatter to the spaces of $x \cdot y > 0$ and $x \cdot y < 0$, respectively. The forward scattering in Figure 1(b) indicates that there is a large angular shift of $20^\circ \pm 2^\circ$ between LCP and RCP patterns, and this results in a four-piece pattern in S_3 component, where $S_3 = I_{LCP} - I_{RCP}$ in this work. This angular shift is far larger than the previously reported values in reflection[14-16], and can be directly seen by eyes or measured by a common system. Therefore, the observed SOI effect in Mie scattering is universal and giant, and own a comprehensive influence in a wide range of fields.

Fig. 1. Universal SOI in Mie scattering. (a) SOI effects from a Mie NS, including the spin-locked twists of the streamline of orbital momentum M_o in near-field, and large angular shift in far-field under x-polarized incident light. (b-c) Calculated forward scattering patterns of S_0 , S_3 , LCP and RCP at $\lambda = 200$ nm (b) and 865 nm (c). The NS with diameter 80 nm is placed in vacuum.

The SOI effect in Mie scattering also distinguishes intrinsically with that from Rayleigh scattering, which is usually treated as near-ideal dipoles[6,20,24]. The angular shift observed in this work disappears when the wavelength is far larger than the NS diameter, which excludes the Rayleigh scattering case [32,33]. And control simulations using an ideal dipole (Fig. S8) also show no angular shifts at any wavelength. However, we note that ideal dipoles

exhibit a strong SOI in their evanescent field according to vectorial angular spectrum theory, and this SOI can transfer into the far field to form a hollow four-leaf scattering pattern in S_3 (Figs. S9-10). This phenomenon, however, requires a wavevector matching condition, and only emits to high angles[20]. The angular shift measured in this work occurs in both backward scattering and at small angles (Fig. S11), where the wavevector matching condition is not satisfied. Thus, the observed SOI in Mie scattering is a new effect.

The interaction between light and NSs is in fact very complex. The transient charge and current distributions differ highly from the ideal dipole (S12 and Movie S1), and both charge and current distributions change over time and position, which makes analysis of the observed SOI effect subtle. To unambiguously reveal this microscopic electrodynamic mechanism, we construct one universal SOI model that includes the conversion of photonic momenta to NS to generate torques in the excitation process and the depolarization of NS to emit photons of specific spin and orbital momenta in the emission process. The continuity equation of photonic spin and orbital angular momenta in an inhomogeneous medium is derived to rigorously describe the complete electrodynamic process. In general, the continuity equation can be simplified as

$$\begin{aligned}\partial_t S + \nabla \cdot \vec{\Sigma} &= -\tau^{\text{spin}} + \Delta^{\text{conversion}} \\ \partial_t L + \nabla \cdot \vec{\Lambda} &= -\tau^{\text{orbit}} - \Delta^{\text{conversion}}\end{aligned}$$

where $\partial_t S$ and $\partial_t L$ describe the angular momentum variation, $\nabla \cdot \vec{\Sigma}$ and $\nabla \cdot \vec{\Lambda}$ refer to the angular momentum flux, τ^{spin} and τ^{orbit} are torques denoting the interaction between photons and matter, and $\Delta^{\text{conversion}}$ is the conversion between spin (SAM) and orbital angular momentum (OAM) [34,35]. The continuity equations of SAM and OAM are highly similar in form.

To calculate the SAM and OAM continuity equations in an inhomogeneous and neutral medium, we adopt the $\mathbf{E}_\perp = -\nabla\varphi$ as the longitudinal electric field, where $\nabla^2\varphi = -\rho_P/\epsilon_0$, and ρ_P is the polarized charge. Similarly, the inhomogeneous magnetization leads to a nonzero polarized magnetic charge, we adopt $\mathbf{H}_\perp = -\nabla\phi$ as the longitudinal magnetic field, where $\nabla^2\phi = -\rho_M$, and ρ_M is the polarized magnetic charge. Then, the total field becomes $\mathbf{E} = -\partial_t \mathbf{A} - \nabla\varphi$ and $\mathbf{H} = -\partial_t \mathbf{C} - \nabla\phi$. Under the Coulomb gauge ($\nabla \cdot \mathbf{A} = 0, \nabla \cdot \mathbf{C} = 0$), we have

$$\begin{aligned}S &= \frac{1}{2}(\epsilon_0 \mathbf{E} \times \mathbf{A} + \mu_0 \mathbf{H} \times \mathbf{C} - \mathbf{P} \times \mathbf{A} + \mu_0 \mathbf{M} \times \mathbf{C}), \\ \vec{\Sigma} &= \frac{1}{2}[(\mathbf{A} \cdot \mathbf{H} - \mathbf{A} \cdot \nabla\phi - \mathbf{C} \cdot \mathbf{E} + \mathbf{C} \cdot \nabla\varphi)\vec{I} - \mathbf{H} \otimes \mathbf{A} + \mathbf{A} \otimes \nabla\phi + \mathbf{E} \otimes \mathbf{C} - \mathbf{C} \otimes \nabla\varphi] \\ , \\ \tau^{\text{spin}} &= \partial_t (\mathbf{P} \times \mathbf{A}) + \mathbf{P} \times \mathbf{E} + \mu_0 \mathbf{M} \times \mathbf{H}, \\ L &= \frac{1}{2}[\epsilon_0 \mathbf{E} \cdot (\mathbf{r} \times \nabla) \mathbf{A} + \mu_0 \mathbf{H} \cdot (\mathbf{r} \times \nabla) \mathbf{C} - \mathbf{P} \cdot (\mathbf{r} \times \nabla) \mathbf{A} + \mu_0 \mathbf{M} \cdot (\mathbf{r} \times \nabla) \mathbf{C}], \\ \vec{\Lambda} &= \frac{1}{2}[\mathbf{H} \times (\mathbf{r} \times \nabla) \mathbf{A} - \mathbf{A} \times (\mathbf{r} \times \nabla) \nabla\phi - \mathbf{E} \times (\mathbf{r} \times \nabla) \mathbf{C} + \mathbf{C} \times (\mathbf{r} \times \nabla) \nabla\varphi], \\ \tau^{\text{orbit}} &= \frac{1}{2}\left\{2\partial_t [\mathbf{P} \cdot (\mathbf{r} \times \nabla) \mathbf{A}] + \mathbf{P} \cdot (\mathbf{r} \times \nabla) \mathbf{E} - \mathbf{E} \cdot (\mathbf{r} \times \nabla) \mathbf{P}\right\} \\ &\quad + \frac{1}{2}\{\mu_0 \mathbf{M} \cdot (\mathbf{r} \times \nabla) \mathbf{H} - \mu_0 \mathbf{H} \cdot (\mathbf{r} \times \nabla) \mathbf{M}\}, \\ \Delta^{\text{conversion}} &= \frac{1}{2}\nabla \cdot (\mathbf{A} \otimes \mathbf{H} - \nabla\phi \otimes \mathbf{A} - \mathbf{C} \otimes \mathbf{E} + \nabla\varphi \otimes \mathbf{C}) + \nabla^2\phi \mathbf{A} - \nabla^2\varphi \mathbf{C} \\ ,\end{aligned}$$

where $\mathbf{A}$ is the magnetic vector-potential, and $\mathbf{C}$ is the electric vector-potential. Obviously, the isotropic but inhomogeneous medium greatly complicates the formula in comparison to the homogeneous case [34,35]. And the detailed derivation process will be reported in other work.

To analyze the interaction between light and NS, we first investigate the electric field $\mathbf{E}$, the polarizability $\mathbf{P}$, and their interaction τ^{spin} inside the NS. Although the NS is neutral, the free electrons inside collectively oscillate under the incident electric field to form a transient induced polarizability $\mathbf{P}$ at different locations in the NS. This induced polarizability $\mathbf{P}$ varies with both the incident electric field $\mathbf{E}$ and the location, and depends highly on the geometry and size of NS. This geometrical influence results in twisting charge and current distributions (Figs. S12 and Movie S1), which differs highly from an ideal dipole. The interaction between the local field and local polarizability $\mathbf{P}$ leads to a local torque on the NS. Because $\mu_0 \mathbf{M} \times \mathbf{H} = 0$ when $\mu_r = 1$, the spin torque τ^{spin} is dominated by the interaction between the magnetic vector-potential $\mathbf{A}$ and polarizability $\mathbf{P}$, as $\tau_1^{\text{spin}} = \partial_t(\mathbf{P} \times \mathbf{A}) = \partial_t \mathbf{P} \times \mathbf{A} - \partial_t \mathbf{A} \times \mathbf{P}$, and the interaction between polarizability $\mathbf{P}$ and electric field $\mathbf{E}$, as $\tau_2^{\text{spin}} = \mathbf{P} \times \mathbf{E}$. From the perspective of averaging time, the influence of τ_1^{spin} vanishes, and the SAM evolution is only governed by the electric interaction τ_2^{spin} (Fig. 2a). Notably, we find a highly-symmetrical torque distribution (Fig. 2a, S13, Movie S2), where the incident linearly-polarized electromagnetic field breaks the symmetry along x and y axes, and leads to a four-piece distribution that is similar to the scattering pattern of S_3 .

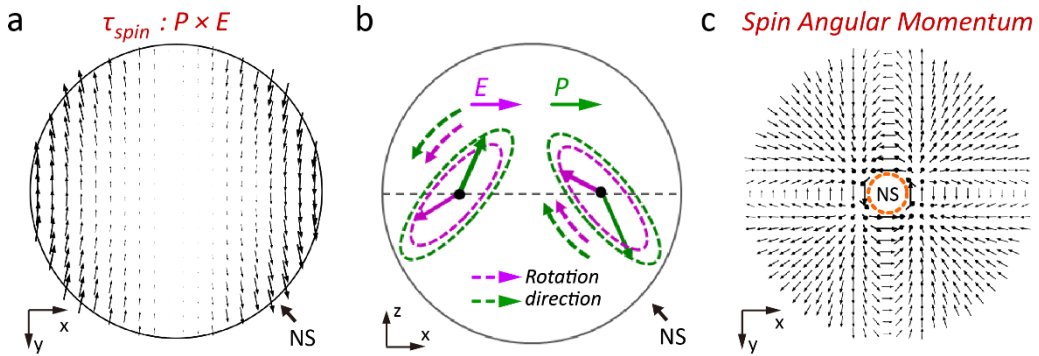


Fig. 2. Influence of SAM on Mie NS. (a) Time-averaged spin torque inside NS for $\tau_2^{\text{spin}} = \mathbf{P} \times \mathbf{E}$. Maximum torque is $\sim 3.6 \times 10^{-12} \text{ Nm}^2$ for 1 Vm^{-1} incident field. (b) Electric field (purple) and induced polarizability (green) inside NS. (c) Calculated SAM density in scattering field outside NS (log scale from $\sim 5 \times 10^{-33}$ to $\sim 5 \times 10^{-28} \text{ kgm}^{-1}\text{s}^{-1}$).

The generated torque drives the polarizability in an elliptical trajectory, which emit light of specific spin in specific directions, and causes the NS depolarization. The spin torque τ^{spin} along $+y$ and $-y$ directions leads to both electric field $\mathbf{E}$ and polarizability $\mathbf{P}$ rotating in the xz plane anticlockwise and clockwise, respectively (Figs. 2b, S14, Movies S3-4). We note that the spinning polarizability $\mathbf{P}$ operates like a circular dipole that prefers to emit LCP light into the direction of τ^{spin} , and RCP into the opposite direction of τ^{spin} (see Fig. S15 and Ref.[36]). This provides an intuitive connection between the four-piece scattering pattern of S_3 and the time-averaged torque distribution in Fig. 2a. These results coincide well with the SAM of the scattering field outside NS: There is an outward SAM in the first and third quadrants, while an inward SAM in the second and fourth quadrants (Fig. 2c).

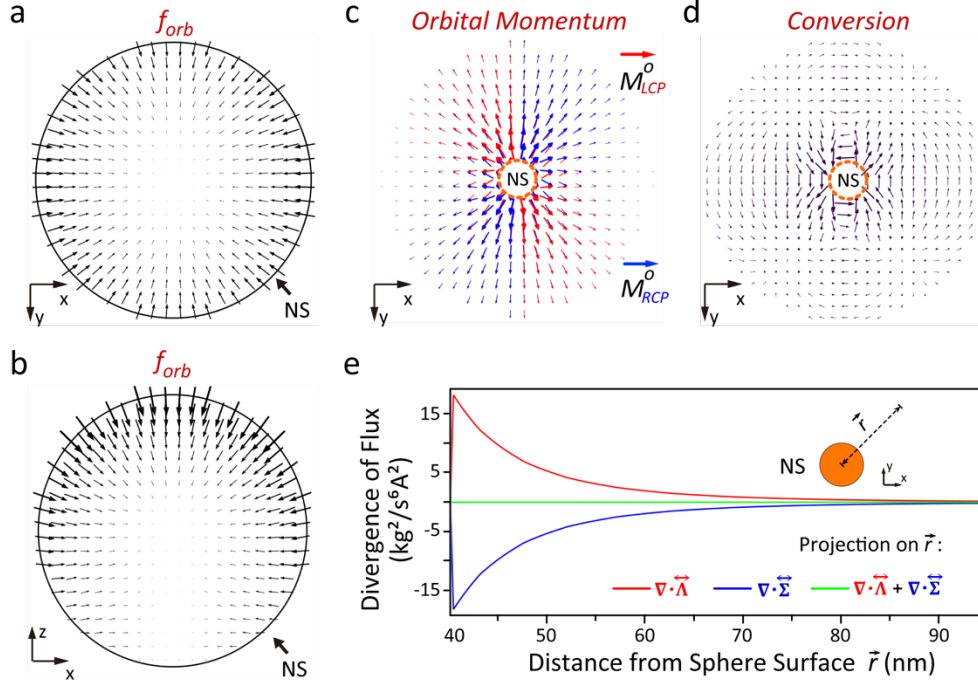


Fig. 3. Influence of OAM on Mie NS. (a-b) Time-averaged orbital force inside NS projected onto the x-y (a) and x-z plane (b). Maximum force is $\sim 6.8 \times 10^{-5} \text{ Nm}^{-3}$ for 1 Vm^{-1} incident field. (c) Calculated LCP (red) and RCP (blue) orbital momenta in scattering field outside NS (log scale from 0 to $\sim 2.4 \times 10^{-21} \text{ kgm}^{-2}\text{s}^{-1}$). (d) Spin-Orbit conversion outside NS (log scale from $\sim 1.2 \times 10^{-15}$ to $\sim 3.5 \times 10^{-12} \text{ Nm}^{-2}$). (e) Calculated projection of $\nabla \cdot \vec{\Sigma}$ and $\nabla \cdot \vec{\Lambda}$ in the scattering field on the direction $\vec{r}$ in xz plane.

In contrast to spin, the orbit-related parts describe the influence from non-uniform physical quantities in space, such as $(\nabla)\mathbf{A}$, $(\nabla)\mathbf{E}$, $(\nabla)\mathbf{P}$ and others in OAM continuity equations. After inducing the polarizability $\mathbf{P}$ inside NS, the spatial non-uniformity generates orbit torque τ^{orbit} on NS, which also participate in driving polarizability $\mathbf{P}$ to rotate. From the orbital momentum density $\mathbf{M}^o$ distributions extracted from $\mathbf{L}$, we see clearly that the scattered LCP light owns larger $\mathbf{M}^o$ in the first and third quadrants, while smaller $\mathbf{M}^o$ in the second and fourth quadrants (Fig. 3a). This further explains the origination of four-leaf pattern in S_3 . When ignoring the position vector $\mathbf{r}$ in τ^{orbit} , we have the orbit force $f^{\text{orbit}} = \frac{1}{2} \left\{ 2 \partial_t [\mathbf{P} \cdot (\nabla)\mathbf{A}] + \mathbf{P} \cdot (\nabla)\mathbf{E} - \mathbf{E} \cdot (\nabla)\mathbf{P} \right\} + \frac{1}{2} \{ \mu_0 \mathbf{M} \cdot (\nabla)\mathbf{H} - \mu_0 \mathbf{H} \cdot (\nabla)\mathbf{M} \}$, where the magnetic contribution can be negligible, and the first term $f_1^{\text{orbit}} = 2 \partial_t [\mathbf{P} \cdot (\nabla)\mathbf{A}]$ vanishes under time-averaging condition. The time-averaging orbit force is thus dominated by $f_2^{\text{orb}} = \mathbf{P} \cdot (\nabla)\mathbf{E}$ and $f_3^{\text{orbit}} = -\mathbf{E} \cdot (\nabla)\mathbf{P}$, which originate from the inhomogeneous electric field and the nonuniform polarization (see Fig. 3a-b, S16, Movie S5). Obviously, the sum of orbit force inside NS cancels in xy plane, but still remains in z-direction. This indicates that the orbit momentum can push the NS going forwards.

According to the continuity equation, the generation of torque, or the depolarization in light emission, always accompanies absorption, or generation of SAM and OAM, respectively. The drastic variation of local wavevector, accompanied by a geometric phase shift, causes strong inter-conversion between SAM and OAM in the near-field, while in fields far away from the NS this inter-conversion is zero (Figs. 3d-e, S17, Movie S6). For the stable system investigated in this work, the time-averaged conversion keeps unchanged (Figs. 3d-e). Because the

absolute values of $\nabla \cdot \vec{\Sigma}$ and $\nabla \cdot \vec{\Lambda}$ are equals to each other (Figs. 3d), the variations in SAM and OAM generate spin and orbit torques, i.e., $\partial_t S + \partial_t L = -\tau^{\text{spin}} - \tau^{\text{orbit}}$, where the spin torque mainly drive the polarizability $\mathbf{P}$ to rotate, while the orbit force pushes the NS going forwards.

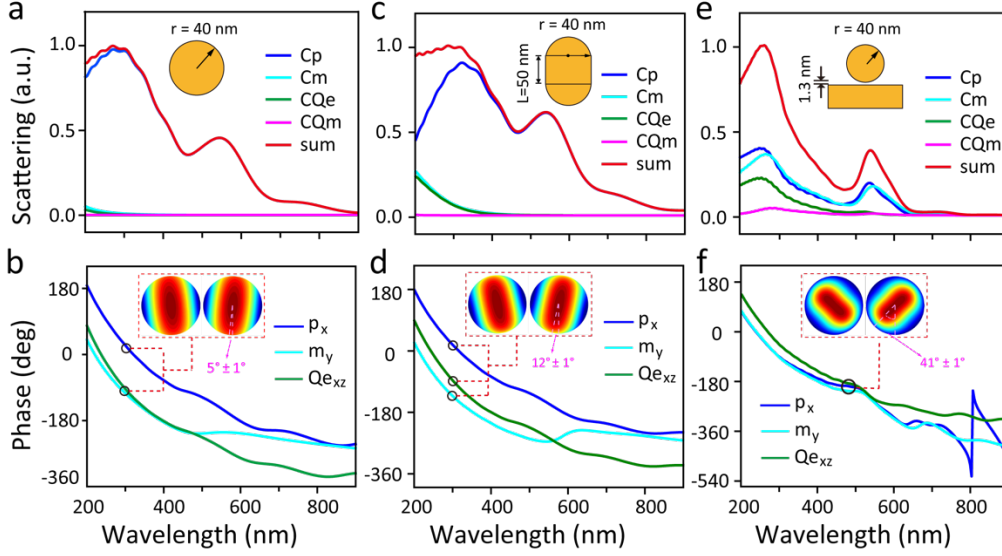


Fig. 4. Multipole superposition view. Scattering intensity and phase of different equivalent dipoles extracted from NS (a-b), nano-rod (c-d), and NS on gold film (e-f). C_p , C_m , C_{Qe} , and C_{Qm} denote

the scattering intensity, where $C_p \propto k^4 |p|^2$, $C_m \propto k^4 c^{-2} |m|^2$, $C_{Qe} \propto \frac{k^4 |Q_e|^2}{120}$, and $C_{Qm} \propto k^6 c^{-2} |Q_m|^2 / 120$. c and k are the light speed and wavevector, respectively. p_x , m_y and $Q_{e_{xz}}$ denote the dipoles in x, y and xz components, respectively. The geometrical models and the LCP and RCP scattering patterns are shown inset.

The SOI effect in Mie scattering also can be simply described by the radiation superposition of multipole. In fact, the retardation effect always makes twisted currents on the spherical surface, where the charge distribution mostly resembles a magnetic dipole, but sporadically also as an electric dipole. Changing the photon energy, geometry, size and materials can effectively modify this retardation effect, as well as the spatiotemporal charge and current distributions, and finally the LCP and RCP scattering patterns. As shown in Fig. 4, we expand the current distribution of NS into a series of multipolar moments based on the multipolar decomposition theory[37], where we see the electric and magnetic dipoles (p and m), and electric and magnetic quadrupoles (Q_e and Q_m). The angular shift can be effectively controlled by changing the intensity and phase of these equivalent dipoles. As indicated by Figs. S18-19, changing the relative intensity or phase between the electric and magnetic dipoles as example can effectively modify the angular shift in the far-field emission. For nanostructures, the equivalent dipoles are controlled by their geometry and size. As shown in Figs. 4(c-d), when we stretch the NS to a nano-rod along the z axis, the intensity of magnetic dipole and electric quadrupoles has been greatly enhanced, leading to a large angular shift of $24^\circ \pm 2^\circ$. Putting NS on metal film can further enhance the intensity of higher order multipoles. As shown in Figs. 4(e-f), the intensity of magnetic dipole is comparable with that of the electric dipole, and the angular shift of $82^\circ \pm 2^\circ$ is observed at 460 nm. These results provide one effective method to modify both near-field and far-field SOI effects in Mie scattering.

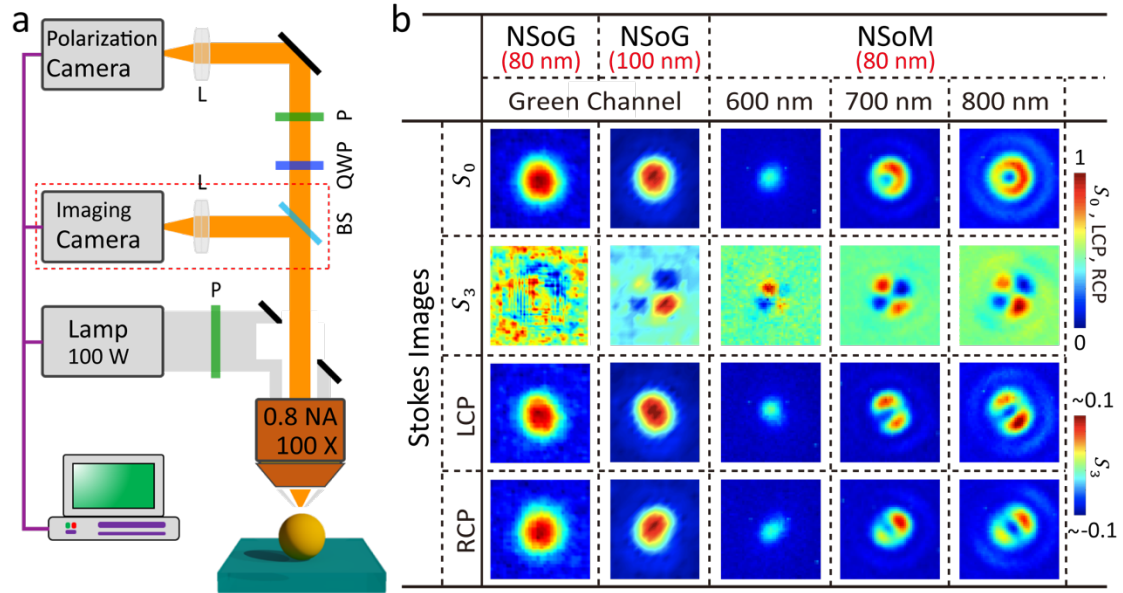


Fig. 5. Experimental verification of giant photonic SHE in nanostructures. (a) Schematic of polarization imaging system, using white light high-angle illumination, lens (L), quarter wave-plate (QWP), beam splitter (BS), and polarizer (P). (b) Scattering patterns of S_0 , S_3 , LCP and RCP in the color camera green channel (480-600 nm), and at filtered wavelengths of 600 nm, 700 nm, and 800 nm, from single nanosphere-on-glass (NSoG) or nanosphere-on-mirror (NSoM) constructs, with NS diameters of 80 nm and 100 nm.

We finally verify experimentally the SOI effect from single nanoparticle scattering by using a carefully-calibrated polarization imaging system (Fig. 5a), where linearly-polarized light illuminates single NPs through a dark-field (DF) objective. Real-space scattering patterns are recorded by an imaging camera (Infinity) to give intensity (S_0) patterns, while a polarization camera following a quarter wave-plate (QWP) and polarizer (P) gives LCP and RCP images (I_{LCP} , I_{RCP}), and thus S_3 patterns. To improve the polarization precision, the beam-splitter (BS) is removed from the optical path before collecting LCP and RCP images. A significant angular shift between LCP and RCP patterns (Fig. 2b) is confirmed in the green channel (480-600 nm) from 100 nm NSs placed on glass (NSoG), which is much weaker for 80 nm NSoGs. Similarly, while the smaller NS shows minimal S_3 , for the 100 nm NSoG a distinct four-leaf S_3 pattern is seen. This phenomenon is even stronger in scattering patterns from 80 nm NSs on gold film (NSoM), where striking four-leaf S_3 patterns are observed at 600 ± 10 nm, 700 ± 10 nm and 800 ± 25 nm (further experimental results in SI Figs. S20-22). These experimental results unambiguously demonstrate a giant SOI effect in Mie scattering that is easily detected even with eyes.

To conclude, we have verified a ubiquitous and significant SOI effect in Mie scattering, which inherently differs with previously-reported phenomenon from Mie NS and Rayleigh scattering. Through constructing the microscopic electrodynamic model, we systematically analyze the whole process of the generation of polarity and torque through absorbing angular momentum, the conversion between SAM and OAM, and the emission of spin photons through depolarization, which explain the origination of SOI in both near-field and far-field. With the multipolar decomposition theory, we also make a simple model to explain and even modify the SOI effect in far-field. These findings apply to all inhomogeneous medium in Mie region, and provide a new perspective to understand and control the polarization and trajectory phenomena in nanophotonics, plasmonics, and beyonds. While the established microscopic electrodynamic model can also works in all of the SOI phenomenon within the framework of Maxwell's theory, including the typical IF and GH

effects [4,23,38], T-spin [38,39], and optical (lateral) forces [40-43]. These findings may inspire diverse applications, including nanophotonic devices, plasmonic circuits, directional chiral emission, etc.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENT

The authors acknowledge financial support from Cambridge Display Technology Ltd., the European Research Council (ERC) under the Horizon 2020 Research and Innovation Programme PICOFORCE (883703), UK EPSRC EP/X037770/1 and EP/L027151/1, the National Natural Science Foundation of China (11604227, 62175222, 62105369), the International Visiting Program for Excellent Young Scholars of SCU (20181504), the Natural Science Foundation of Hunan Province (2025JJ40065), and the International Science and Technology Innovation Cooperation of Sichuan Province (2021YFH0137, 2025YFHZ0239).

[*Science 340, 328-330 \(2013\)*](#)

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